

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

PHYSICS | SSC-I (9th Grade) | Units 1-6 and 9 | Marking Report

Exam: exam-09-ch1-6-9 (Version 1) Candidate: Seemab Date: 7 July 2026 Total Marks: 65

Deep-check verified: independently re-marked from scratch, page by page; MCQ 12/12 confirmed letter by letter and every Section C calculation re-derived (screw gauge 4.32 mm, collision 10.71 m/s, orbital speed 6981 m/s + centripetal 3046 N, $P=\rho gh \rightarrow 60$ kPa). Both markings agree on **64.5/65 (99.2%)**.

64.5 / 65 99.2% A+

Section A: 12/12 | Section B: 32.5/33 | Section C: 20/20 | MCQ: perfect (streak restarts at 1)

SECTION A: MCQS

12/12

100% (Perfect)

SECTION B: SHORT

32.5/33

98.5% (1 half-deduction)

SECTION C: LONG

20/20

100% (all 4 correct)

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question (abbreviated)	Seemab	Correct	Result
1	Significant figures in 0.00650 s	A	A (3)	CORRECT
2	Least count of a vernier caliper (1 mm, 20 divisions)	B	B (0.05 mm)	CORRECT
3	Displacement of athlete returning to start	C	C (Zero)	CORRECT
4	Scooter acceleration 4→22 m/s in 6 s	D	D (3 m/s ²)	CORRECT
5	Swimmer pushes water, water pushes back	C	C (Newton's 3rd law)	CORRECT
6	Momentum of 300 kg rickshaw at 6 m/s	C	C (1800 kg·m/s)	CORRECT
7	Rotational (2nd) equilibrium requires zero...	B	B (Net torque)	CORRECT
8	Centripetal force on stone (0.2 kg, 0.5 m, 4 m/s)	A	A (6.4 N)	CORRECT
9	Spring constant (8 cm stretch, 16 N)	D	D (200 N/m)	CORRECT
10	Average atmospheric pressure at sea level	C	C (760 mmHg)	CORRECT
11	GPE of 3 kg lifted to 8 m	D	D (240 J)	CORRECT
12	Branch studying fundamental particles	B	B (Particle physics)	CORRECT

MCQ score: **12/12 (Perfect)**. **Perfect-MCQ streak restarts** after it ended yesterday on Chemistry exam-14 (11/12). Physics MCQ stays a flawless 84/84 (100%) across all 7 physics papers.

Section B — Short Answer Questions (11 × 3 = 33 Marks)

All 11 parts attempted. OR variant chosen for 2(ii), 2(vii), 2(viii), 2(xi).

Q.2(i): Scientific Notation & SI Prefixes

Chose: Main 3/3

(a) $0.00072\text{ m} = 7.2 \times 10^{-4}\text{ m}$ ✓ (b) $8,900,000\text{ m} = 8.9 \times 10^6\text{ m}$ ✓ prefix: $7.2 \times 10^{-4}\text{ m} = 0.72 \times 10^{-3}\text{ m} = 0.72\text{ mm (milli)}$ ✓. All correct.

Q.2(ii): Rounding to Three Significant Figures

Chose: OR alternative 3/3

(a) $45.268 \rightarrow 45.3$ ✓ (b) $0.005672 \rightarrow 0.00567$ ✓ (c) $9432 \rightarrow 9430$ ✓. All three correct.

Q.2(iii): Distance & Displacement

Chose: Main 3/3

(a) Total distance = $6 + 8 = 14\text{ km}$ ✓. (b) Displacement via Pythagoras = $\sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100} = 10\text{ km}$ ✓, and she states the direction (north-east of the starting point) ✓. Full marks.

Q.2(iv): Acceleration & Distance Covered

Chose: Main 3/3

(a) $a = (v_f - v_i)/t = (25 - 5)/8 = 2.5\text{ m/s}^2$ ✓. (b) average velocity = $(5+25)/2 = 15\text{ m/s}$; distance = $15 \times 8 = 120\text{ m}$ ✓. (Trivial “m/s²” unit slip, self-corrected to m.)

Q.2(v): Newton's Second Law — Acceleration

Chose: Main 3/3

$a = F/m$: for 4 kg, $a = 18/4 = 4.5\text{ m/s}^2$ ✓; for 12 kg, $a = 18/12 = 1.5\text{ m/s}^2$ ✓. States acceleration is inversely proportional to mass ✓.

Q.2(vi): Impulse & Average Force

Chose: Main3/3

(a) Impulse = $m(v_f - v_i) = 0.2(0 - 15) = -3 \text{ kg}\cdot\text{m/s}$, magnitude $3 \text{ N}\cdot\text{s}$ ✓. (b) $F = \Delta p/\Delta t = -3/0.1 = -30 \text{ N}$, magnitude 30 N ✓. Explains the negative sign (force opposes motion, bringing the ball to rest) ✓.

Q.2(vii): Principle of Moments (Seesaw)

Chose: OR alternative3/3

Clockwise moment = anticlockwise moment: $480 \times 0.25 = 400 \times d \rightarrow 120 = 400d \rightarrow d = 120/400 = 0.3 \text{ m}$ ✓. Bilal sits 0.3 m from the pivot.

Q.2(viii): Average Orbital Speed

Chose: OR alternative3/3

$v = 2\pi r/T = (2 \times 3.14 \times 7,000,000)/6000 = 7330 \text{ m/s} = 7.33 \text{ km/s}$ ✓. (Intermediate product slightly miswritten but the final answer matches the key.)

Q.2(ix): Spring Constant (Hooke's Law)

Chose: Main3/3

Force stretching spring = weight = $mg = 3 \times 10 = 30 \text{ N}$ ✓. By Hooke's law, $k = F/x = 30/0.06 = 500 \text{ N/m}$ ✓.

Q.2(x): Kinetic Energy

Chose: Main2.5/3

$KE_1 = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 12^2 = 144 \text{ J}$ ✓. When speed doubles to 24 m/s , $KE_2 = \frac{1}{2} \times 2 \times 24^2 = 576 \text{ J}$ ✓. Both calculations correct. **-0.5**: the required conclusion — that doubling the speed makes KE **four times** greater because $KE \propto v^2$ ($576 = 4 \times 144$) — was not stated. The ratio is sitting in her own two numbers; she just needed to name it.

Q.2(xi): Fields of Physics

Chose: OR alternative3/3

Three fields defined with examples: biophysics (blood-flow fluid dynamics), astrophysics (formation of stars), acoustics (concert-hall sound). All three correct — full marks.

Section B total: 32.5 / 33 (98.5%)

Section C — Long Answer Questions (4 × 5 = 20 Marks)

All 4 long questions attempted — OR variant chosen for every one.

Q.3(i): Screw Gauge — Definition, Pitch, Least Count & Diameter

Chose: OR alternative5/5

Screw gauge defined (measures a fraction of the smallest linear-scale division by rotating a circular scale; finer than a vernier caliper) ✓. Pitch = distance the circular scale moves along the main scale in one rotation; least count = $\text{pitch}/\text{divisions} = 1 \text{ mm}/100 = 0.01 \text{ mm}$ ✓. Diameter = $\text{MSR} + (\text{CSR} \times \text{LC}) = 4 + (32 \times 0.01) = 4 + 0.32 = 4.32 \text{ mm}$ ✓. Full marks.

Q.3(ii): Newton's Three Laws & Collision

Chose: OR alternative5/5

Three laws, each with a daily-life example: 1st (book at rest until a force moves it), 2nd (pushing a trolley harder gives greater acceleration), 3rd (swimmer pushes water back, is pushed forward) ✓. Conservation of momentum setup $m_1v_1 + m_2v_2 = (m_1+m_2)v$ with $v_2=0$ ✓. $(2000 \times 15)/(2800) = 30,000/2800 = 10.71 \text{ m/s}$ ✓. Full marks.

Q.3(iii): Centripetal Force & Orbital Motion (Satellite)

Chose: OR alternative5/5

Centripetal force and orbital motion both defined (force directed towards the centre; the repeating circular path a satellite follows under gravity) ✓. (a) $v = 2\pi r/T = (2 \times 3.14 \times 8,000,000)/7200 = 6981 \text{ m/s}$ ✓. (b) $F = mv^2/r = (500 \times 6981^2)/8,000,000 = 3046 \text{ N}$ ✓ (she corrected an initial 3004 to 3046). Full marks.

Q.3(iv): Derive P = ρgh & Water-Tank Pressure

Chose: OR alternative5/5

Full derivation: liquid column of height h , base area A , density ρ ; force on base = weight = mg ; with $m = \rho Ah$, $F = \rho Ahg$; $P = F/A = \rho gh$ ✓. Calculation: $P = \rho gh = 1000 \times 10 \times 6 = 60,000 \text{ Pa} = 60 \text{ kPa}$ ✓. Full marks.

Section C total: 20 / 20 (100%)

Complete Deduction Log (reconciles to headline score)

Question	Awarded	Lost	Reason
Q.2(x) main: kinetic energy	2.5/3	-0.5	$KE_1 = 144 \text{ J}$ and $KE_2 = 576 \text{ J}$ both correct, but the required relationship ($KE \propto v^2$, so doubling the speed makes KE four times greater) was not stated.
Total deductions	-0.5		65 - 0.5 = 64.5/65 confirmed. Every other question full marks.

Strengths in this attempt

- Perfect 12/12 MCQ — restarts the perfect-MCQ run; physics MCQ now 84/84 (100%) across all 7 papers.
- Every Section C long question (all four OR variants) fully correct: screw gauge 4.32 mm, collision 10.71 m/s, orbital speed 6981 m/s + centripetal 3046 N, and the $P=\rho gh$ derivation + 60 kPa.
- Numerical work faultless across the paper: acceleration, impulse and sign, moments, Hooke's law, orbital speed.
- Direction-aware where it matters (displacement north-east; impulse/force negative sign explained).
- Her highest physics score to date and a near-flawless paper.

Areas to revise

- State the relationship, not just the number (-0.5):** when a question asks "what would the new value be if X is doubled," add the reasoning ($KE \propto v^2$, so $4\times$ greater). This one line was the only mark between this and a perfect paper.
- Re-check intermediate arithmetic:** a few intermediate products in the $2\pi r$ calculations were miswritten, even though every final answer landed correctly. A quick glance back at each line avoids a lucky-looking slip.

Physics Exam Comparison

Exam	Date	Score	%	MCQ	Sec B	Sec C	Grade
exam-01-ch1-2	7 Apr 2026	63.5/65	97.7%	12/12	32.5/33	19/20	A+
exam-02-ch3-4-9	10 Apr 2026	62/65	95.4%	12/12	30.5/33	19.5/20	A+
exam-03-ch5	18 May 2026	65/65	100%	12/12	33/33	20/20	A+
exam-04-ch6	22 Jun 2026	55.5/65	85.4%	12/12	29/33	14.5/20	A
exam-07-ch1-6-9	25 Jun 2026	58.5/65	90.0%	12/12	31.5/33	15/20	A+
exam-08-ch1-6-9	2 Jul 2026	61/65	93.8%	12/12	30.5/33	18.5/20	A+
exam-09-ch1-6-9 (this)	7 Jul 2026	64.5/65	99.2%	12/12	32.5/33	20/20	A+

exam-09 (64.5/65, 99.2%) is Seemab’s **highest physics score to date** and her strongest full-syllabus Ch1-6+9 paper by a clear margin (over exam-07’s 90.0% and exam-08’s 93.8%). Physics subject average rises to 94.5% (430.0/455, 7 exams), edging just past Mathematics (94.4%) to become her **strongest subject**. Physics MCQ remains a perfect 84/84 (100%) across all 7 papers.